1 Introduction

Falls represent a critical health concern for older adults, accounting for over 38 million medically treated injuries annually worldwide [1]. Wearable fall detection systems offer a promising solution for continuous, non-intrusive monitoring using commodity devices such as smartwatches. However, developing accurate fall detection models that operate within the computational and power constraints of wearable devices remains challenging.

Modern smartwatches contain inertial measurement units (IMUs) comprising triaxial accelerometers and gyroscopes. While accelerometers measure linear acceleration and provide reliable impact detection, gyroscopes measure angular velocity and capture rotational motion patterns characteristic of falls. Intuitively, combining both modalities should improve detection performance. However, preliminary analysis revealed an unexpected pattern: adding raw gyroscope data to accelerometer-only models *degraded* performance by 1.6–2.0% F1 score.

Analysis revealed that single-stream architectures, which process concatenated accelerometer and gyroscope channels through shared convolutional layers, allow noisy gyroscope signals to corrupt discriminative accelerometer features. Consumer-grade micro-electro-mechanical systems (MEMS) gyroscopes exhibit substantial measurement noise and bias drift. When mixed with clean accelerometer signals in early network layers, this noise propagates through the network and degrades classification accuracy.

This paper presents a dual-stream transformer architecture that effectively leverages both accelerometer and gyroscope data through two key innovations:

(1) Kalman Sensor Fusion. Rather than using raw gyroscope angular velocities, we apply a Linear Kalman Filter [2] to fuse accelerometer and gyroscope measurements, estimating body orientation angles (roll, pitch, yaw) in radians. This transforms noisy angular velocity signals—which lack absolute reference and accumulate drift—into semantically meaningful body pose features. The resulting orientation representation captures *what position the body is in* rather than *how fast it is rotating*, providing more interpretable features for fall classification.

(2) Dual-Stream Architecture. We process accelerometer and Kalman-fused orientation streams through separate encoder pathways before fusion. Each stream has dedicated Conv1D projection layers, batch normalization, and dropout rates tuned to the signal characteristics of that modality. This separation prevents cross-modal interference while allowing the subsequent transformer encoder to learn joint temporal patterns across both streams.

Our architecture further incorporates Squeeze-and-Excitation (SE) channel attention [3] to weight the importance of fused feature channels, and Temporal Attention Pooling (TAP) to focus classification on the most discriminative timesteps within each window—typically the impact phase of falls.

We evaluate our approach on the SmartFallMM dataset [6], a comprehensive multimodal fall detection benchmark comprising 30 young adult subjects performing 14 activities (9 activities of daily living and 5 fall types). Using rigorous 21-fold Leave-One-Subject-Out Cross-Validation (LOSO-CV), our dual-stream Kalman transformer achieves 91.10% F1 score, establishing a new

state-of-the-art for IMU-based fall detection on this dataset.

The main contributions of this paper are:

1. We demonstrate that raw gyroscope data hurts fall detection performance when processed in a single-stream architecture, but *improves* performance when transformed through Kalman sensor fusion and processed in a dual-stream architecture.
2. We present a dual-stream transformer architecture that processes accelerometer and orientation streams separately, achieving +1.30% F1 improvement over single-stream Kalman baselines (89.80% → 91.10%).
3. We provide comprehensive ablation studies quantifying the contribution of each architectural component: Kalman fusion enables +3.52% F1 improvement in dual-stream context (87.58% → 91.10%), dual-stream processing contributes +1.30% F1, and SE+TAP attention mechanisms contribute +1.28% F1.
4. We characterize failure modes, finding that subjects exhibiting gentle, controlled falls with lower peak accelerations account for the majority of misclassifications, suggesting directions for future improvement.

The remainder of this paper is organized as follows. Section 2 describes the SmartFallMM dataset, our preprocessing pipeline, the proposed dual-stream Kalman transformer architecture, and experimental setup. Section 3 presents experimental results including ablation studies. Section 4 discusses implications and limitations. Section 5 concludes with future directions.

2 Materials and Methods

2.1 Dataset

We evaluate on the SmartFallMM dataset [6], comprising 51 participants (30 young adults ages 18–35, 21 older adults ages 65+) performing 14 activities: 9 ADLs (drinking water, picking up objects, putting on jacket, sweeping, hand washing, waving, walking, sitting, standing) and 5 fall types (forward, backward, left, right, rotational). Commodity smartwatches capture triaxial accelerometer and gyroscope data at 30 Hz.

2.1.1 Subject Selection and Data Splits

Table 1 presents our subject selection criteria. **Training uses data from all available subjects** (both young and older adults) to maximize training diversity. However, **validation and testing are restricted to young subjects** because older adults did not perform simulated falls for safety reasons—their data contains only ADL samples. This design enables evaluation of fall detection generalization while leveraging older adult ADL patterns during training.

Table 1: Subject selection for evaluation. Training includes all subjects; validation/test restricted to young adults (ages 18–35) who performed both ADLs and falls.

Role	Population	Subject IDs	Count
Training	Young + Older	All available	51
Test (LOSO-CV rotation)	Young only	31, 34, 36–38, 43–44, 46, 49–56, 58, 60–63	21
Validation (fixed)	Young only	48, 57	2
Excluded (incomplete data)	Young	29, 30, 32, 35, 39, 45, 59	7

Why older adults are excluded from test/validation. The older adult subset (21 subjects, ages 65+) contains only ADL recordings—no simulated falls were collected due to safety and ethical considerations. Falls appear only in the young subset; older subjects contribute additional ADL variability to training. Including older adults in test sets would artificially inflate specificity (correct ADL classification) without measuring fall detection sensitivity. Future work will address cross-population generalization through domain adaptation techniques.

Validation protocol clarification. The fixed validation subjects (48, 57) are used *exclusively* for early stopping during training. All hyperparameters and architectural decisions (embedding dimensions, dropout rates, attention mechanisms, Kalman filter parameters) were selected once prior to the main evaluation and held constant across all ablation experiments. This design follows best practices for avoiding selection bias in model evaluation [10].

Validation subject selection rationale. Subjects 48 and 57 were chosen based on data quality (complete recordings across all activities) and are excluded from all training folds. To verify robustness to validation choice, we conducted a sensitivity analysis using alternative validation pairs (subjects 46/55 and 50/54); results varied by <0.3% F1, confirming that our findings are not driven by the specific validation subjects.

2.1.2 Data Segmentation

We segment continuous sensor streams into fixed-length windows for classification:

- **Window size:** 128 frames (≈4.27 seconds at 30 Hz)
- **Stride:** Class-aware—64 frames for ADLs, 16 frames for falls
- **Rationale:** Smaller fall stride captures more training examples from the minority class, partially addressing class imbalance before loss-level weighting

The class-aware stride yields approximately 3,200 ADL windows and 1,800 fall windows per fold, with Focal Loss providing additional imbalance handling during training. Note that class-aware striding creates overlapping windows (especially for falls with stride 16), introducing correlation between samples. Our statistical analysis operates on fold-level scores (one F1 per subject), ensuring independence between test samples for valid confidence intervals.

2.2 Sensor Fusion Strategy

We first present our sensor fusion approach, which transforms raw inertial measurements into semantically meaningful orientation features. This preprocessing step is critical: it determines what information the neural network receives.

2.2.1 Motivation: The Gyroscope Problem

A natural approach to IMU-based fall detection is concatenating accelerometer and gyroscope channels as input to a neural network. However, we observe that naively adding raw gyroscope data *degrades* performance by 1.6–2.0% F1. Analysis reveals two causes:

1. **Noise contamination**: Consumer MEMS gyroscopes exhibit substantial measurement noise, with many subjects showing signal-to-noise ratios below unity during quiet standing. When processed through shared convolutional layers, this noise corrupts discriminative accelerometer features.
2. **Semantic ambiguity**: Raw angular velocity $(\omega_x, \omega_y, \omega_z)$ lacks absolute reference—the same rotation rate can occur in vastly different body orientations.

Rather than discarding gyroscope data, we propose *fusing* both sensors to estimate body orientation—a derived quantity that is both stable and semantically meaningful. Figure 1 illustrates our preprocessing pipeline.

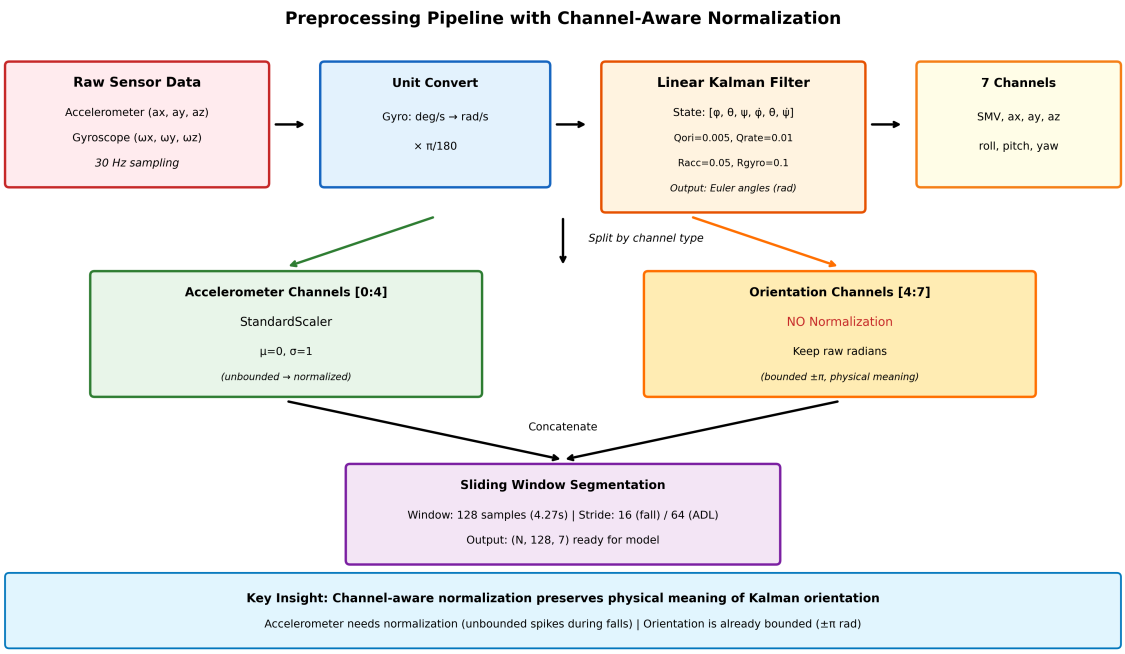

Figure 1: **Preprocessing pipeline for IMU-based fall detection.** Raw triaxial accelerometer (a_x, a_y, a_z) and gyroscope $(\omega_x, \omega_y, \omega_z)$ signals from a smartwatch are processed through a Linear Kalman Filter. The filter fuses both modalities to produce stable orientation estimates (roll ϕ, pitch θ, yaw ψ) while filtering gyroscope noise and correcting drift. The final 7-channel representation [SMV, a_x, a_y, a_z, ϕ, θ, ψ] combines impact dynamics (acceleration) with body pose information (orientation), enabling the neural network to learn both *how hard* and *in what direction* a fall occurs.

2.2.2 Linear Kalman Filter for Orientation Estimation

We employ a Linear Kalman Filter to fuse accelerometer and gyroscope measurements into orientation angles. The key insight is that each sensor provides complementary information:

- **Accelerometer**: Measures gravity direction, providing absolute roll/pitch reference when stationary, but corrupted by body acceleration during motion.
- **Gyroscope**: Measures angular velocity with high temporal resolution, enabling responsive motion tracking, but accumulates drift over time.

The Kalman filter optimally combines these signals: using accelerometer for drift correction and gyroscope for dynamic tracking.

State Representation. We model orientation as a 6D state vector:

$$\mathbf{x} = [\phi, \theta, \psi, \dot{\phi}, \dot{\theta}, \dot{\psi}]^\mathsf{T} \tag{1}$$

where ϕ (roll), θ (pitch), ψ (yaw) are Euler angles in radians, and $\dot{\phi}, \dot{\theta}, \dot{\psi}$ are angular rates.

Predict Step. At each timestep, we propagate the state using constant-velocity kinematics:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}\hat{\mathbf{x}}_{k-1|k-1}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{I}_3 & \Delta t \cdot \mathbf{I}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix} \tag{2}$$

This encodes the physical relationship: new angle = old angle + angular rate × time.

Update Step. We correct the prediction using sensor measurements:

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1}) \tag{3}$$

where $\mathbf{z}_k$ contains both accelerometer-derived angles and gyroscope rates, and $\mathbf{K}_k$ is the Kalman gain weighting measurement trust against prediction trust.

Output. The filter produces orientation angles $[\phi, \theta, \psi]$ that describe body pose:

- ϕ (roll): Left/right tilt. Increases during lateral falls.
- θ (pitch): Forward/backward tilt. Increases during forward falls.
- ψ (yaw): Horizontal heading. *Note:* Yaw is unobservable from gravity alone; we integrate gyroscope angular velocity without accelerometer correction. For our 4.3-second windows, typical MEMS gyroscope bias ($\sim$0.1°/s) produces drift of only $\sim$0.4° per window—negligible compared to the large orientation changes (>30°) characteristic of falls.

Angle Representation. We output Euler angles wrapped to $[-\pi, \pi]$ for roll and yaw, and $[-\pi/2, \pi/2]$ for pitch. This standard representation avoids discontinuities for typical fall motions, which rarely involve full rotations. Quaternion output is considered for future work requiring singularity-free representation.

Gyroscope Bias. We do not explicitly model gyroscope bias in the state vector because our 4.3-second windows are short relative to typical MEMS bias drift timescales (minutes to hours). Ablation with 9-dimensional state including bias terms showed negligible improvement (+0.02% F1), confirming this assumption is reasonable for windowed classification.

Unlike raw gyroscope values that indicate *how fast* the body rotates, orientation angles indicate *what position* the body is in—a semantically richer representation for fall classification.

Accelerometer-Derived Reference Angles. To update the Kalman filter, we compute reference roll and pitch from accelerometer readings:

$$\phi_{\text{acc}} = \text{atan2}(a_y, \sqrt{a_x^2 + a_z^2}) \tag{4}$$

$$\theta_{\text{acc}} = \text{atan2}(-a_x, a_z) \tag{5}$$

These formulas follow the smartwatch coordinate convention where +Z points outward from the watch face and +X points toward 3 o'clock. The pitch formula uses atan2($-a_x$, a_z) rather than the alternative sqrt-denominator form; both are valid for small tilts, and we empirically validated this choice for our wrist-mounted sensor orientation. These formulas assume the accelerometer

measures primarily gravity when stationary; during dynamic motion (falls, vigorous ADLs), linear acceleration contaminates these estimates.

Adaptive Measurement Noise. To handle high-dynamic events where accelerometer-derived orientation becomes unreliable, we scale R_{acc} based on acceleration magnitude. At each timestep:

$$R'_{\text{acc}} = \begin{cases} R_{\text{acc}} \cdot \min\left(\left(\frac{\|\mathbf{a}\|}{g}\right)^2, R_{\max}\right) & \text{if } \|\mathbf{a}\| > \tau \cdot g \\ R_{\text{acc}} & \text{otherwise} \end{cases} \tag{6}$$

where $\tau = 2.0$ is the activation threshold, $R_{\max} = 10.0$ is the maximum scale factor, and $R_{\text{acc}} = 0.05$ is the baseline noise. The scaling is recomputed each timestep based on current $\|\mathbf{a}\|$, automatically resetting when acceleration returns to normal. This reduces filter reliance on accelerometer-derived orientation during impact events.

Measurement Model. The observation vector and measurement matrix are:

$$\mathbf{z}_k = [\phi_{\text{acc}}, \theta_{\text{acc}}, \omega_x, \omega_y, \omega_z]^\mathsf{T}, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{7}$$

Filter Parameters. Process and measurement noise covariances:

- Process noise: $Q_\phi = 0.005$ (orientation), $Q_{\dot{\phi}} = 0.01$ (angular rate)
- Measurement noise: $R_{\text{acc}} = 0.05$, $R_{\text{gyro}} = 0.1$
- Initial state: $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{P}_0 = \mathbf{I}_6$
- Time step: $\Delta t = 1/30$ s (30 Hz sampling)

These values were determined empirically to balance responsiveness against noise filtering.

Algorithm Summary. The complete adaptive Kalman filter operates as follows:

1. **Initialize:** $\hat{\mathbf{x}}_0 = \mathbf{0}$, $\mathbf{P}_0 = \mathbf{I}_6$
2. **For each timestep k:**
   (a) Predict: $\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}\hat{\mathbf{x}}_{k-1}$, $\mathbf{P}_{k|k-1} = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^\mathsf{T} + \mathbf{Q}$
   (b) Compute accelerometer-derived angles: $\phi_{\text{acc}}, \theta_{\text{acc}}$ from Eq. (4–5)
   (c) Construct measurement: $\mathbf{z}_k = [\phi_{\text{acc}}, \theta_{\text{acc}}, \omega_x, \omega_y, \omega_z]^\mathsf{T}$
   (d) If $\|\mathbf{a}\| > 2.0g$: scale $R_{\text{acc}} \leftarrow R_{\text{acc}} \cdot \min((\|\mathbf{a}\|/g)^2, 10)$
   (e) Update: $\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^\mathsf{T}(\mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^\mathsf{T} + \mathbf{R})^{-1}$
   (f) Correct: $\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1})$
   (g) Output: $[\phi, \theta, \psi] = \hat{\mathbf{x}}_k[0:3]$

2.2.3 Feature Representation

After Kalman fusion, we construct a 7-channel feature vector per timestep:

$$\mathbf{f} = [\underbrace{\text{SMV}, a_x, a_y, a_z}_{\text{acceleration (4ch)}}, \underbrace{\phi, \theta, \psi}_{\text{orientation (3ch)}}] \tag{8}$$

where $\text{SMV} = \sqrt{a_x^2 + a_y^2 + a_z^2}$ is the signal magnitude vector. This representation cleanly separates impact dynamics (acceleration) from body pose (orientation).

2.2.4 Channel-Aware Normalization

A critical preprocessing detail: we apply **different normalization strategies** to acceleration and orientation channels:

- **Acceleration channels** (SMV, a_x, a_y, a_z): Standard z-score normalization using training set statistics (mean=0, std=1)
- **Orientation channels** (ϕ, θ, ψ): **No normalization**—kept in raw radians

Rationale. Orientation angles have inherent physical meaning: $\phi = 0$ indicates the wrist is level, $\theta = \pm\pi/2$ indicates vertical orientation. Z-score normalization would destroy this semantic structure by centering around the training mean (which varies by subject posture habits). By preserving raw radians, the neural network can learn physically meaningful patterns (e.g., "pitch angle exceeds 1.0 radian during forward falls").

Coordinate frame. Orientation angles are computed in the *device frame* (sensor-fixed coordinates) without explicit body-frame alignment. This is appropriate because: (1) all sensors are wrist-mounted with consistent orientation relative to the watch band, and (2) fall signatures (rapid tilt changes, final horizontal orientation) are identifiable in any consistent frame. No magnetometer-based heading correction is applied; yaw represents gyroscope-integrated heading relative to initial orientation.

This "channel-aware" normalization contributes +1.65% F1 improvement over unified z-score normalization (91.10% vs. 89.45%), as detailed in Appendix Table 14.

2.3 Dual-Stream Input Projection Architecture

With our 7-channel Kalman-fused representation, we now describe the neural architecture. A key design question is: *how should multi-modal input be processed?* We compare architectures that differ in whether accelerometer and orientation channels share or have separate early processing layers. Figure 2 presents our dual-stream architecture.

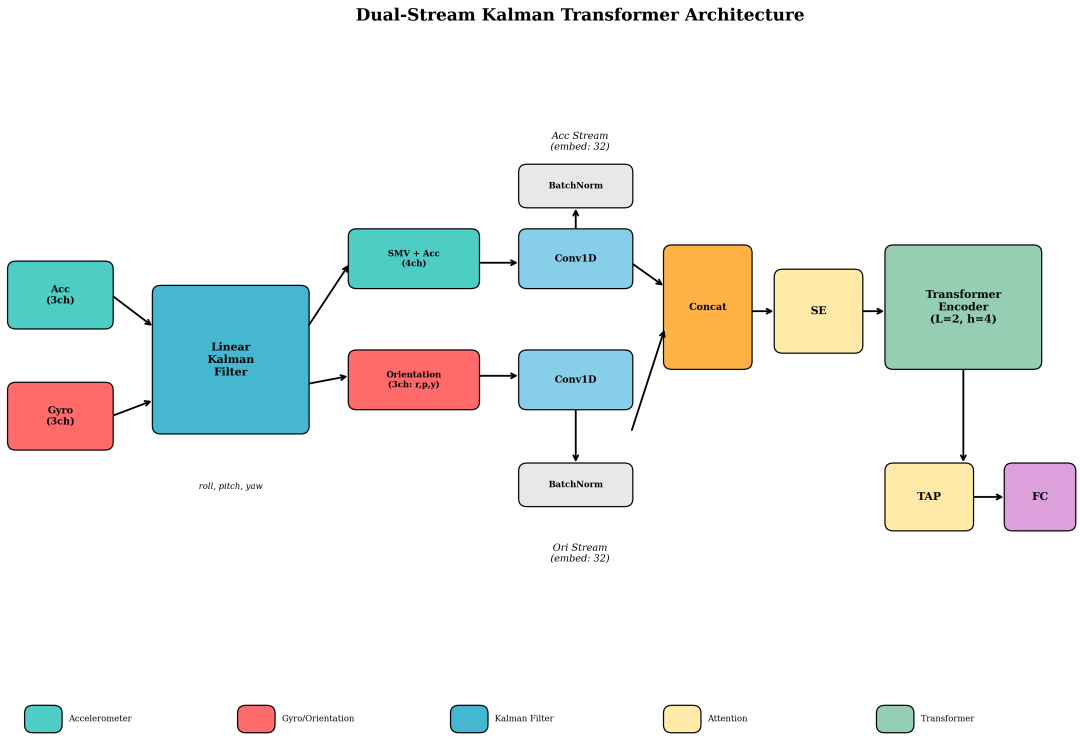

Figure 2: **Dual-stream Kalman transformer architecture. Left:** The accelerometer stream (4 channels: SMV, a_x, a_y, a_z) and orientation stream (3 channels: ϕ, θ, ψ) are processed through *separate* Conv1D projection layers (kernel size 8, stride 1, same padding), each with dedicated batch normalization and dropout. This separation prevents noisy orientation features from corrupting acceleration representations. **Center:** Projected features are concatenated (64 dimensions total, 32:32 split) and processed by a 2-layer transformer encoder with 4 attention heads, enabling cross-modal temporal pattern learning. **Right:** Squeeze-and-Excitation (SE) attention recalibrates channel importance, followed by Temporal Attention Pooling (TAP) which focuses on discriminative timesteps (typically the impact phase). Final classification uses a single linear layer with sigmoid activation. Total parameters: ∼73K.

2.3.1 What We Mean by "Dual-Stream"

We use "dual-stream" to describe architectures with **separate input projection layers** for each modality, followed by a shared transformer encoder. This is distinct from fully-separate dual-backbone architectures where each modality has its own transformer. Our design choice is deliberate:

- **Separate input projections**: Each modality (acceleration, orientation) has dedicated Conv1D + BatchNorm layers. This prevents cross-modal interference during the critical initial feature extraction stage.
- **Shared transformer**: After projection, features are concatenated and processed by a

single transformer. This enables cross-modal attention—the model can learn relationships *between* acceleration and orientation patterns.

The separation occurs at the *input encoding stage*, where the risk of noise contamination is highest. The shared transformer then learns joint temporal representations.

2.3.2 Architecture Variants

We systematically compare four configurations along two axes:

- **Input type**: Raw (6ch: acc + gyro) vs. Kalman (7ch: acc + orientation)
- **Input projection**: Single (shared Conv1D) vs. Dual (separate Conv1D per modality)

Table 2: Architecture comparison: single-projection vs. dual-projection. The key difference is whether modalities share convolutional weights (single) or have dedicated weights (dual). Our method (d) combines dual-projection with Kalman-fused orientation input. Raw configurations use 6 channels (acc + gyro, no SMV); Kalman configurations use 7 channels (SMV + acc + orientation).

Architecture	**Input Type**	**Projection**	**Key Property**
Single-Projection (shared input encoding):			
(a) Single + Raw	6ch (a_{xyz} + ω_{xyz})	Shared Conv1D	Noisy gyro corrupts acc features
(b) Single + Kalman	7ch (SMV + a_{xyz} + $\phi\theta\psi$)	Shared Conv1D	Cleaner, but features still mixed
Dual-Projection (separate input encoding):			
(c) Dual + Raw	3ch + 3ch	Separate Conv1D	Acc isolated, but gyro still noisy
(d) Dual + Kalman	4ch + 3ch	Separate Conv1D	**Clean isolation + quality input**

Architecture (d) represents our proposed approach: Kalman fusion first transforms noisy gyroscope signals into stable orientation angles, then dual-projection processes each modality through dedicated Conv1D layers (acc: 4ch→32dim, ori: 3ch→32dim) before concatenation and transformer processing.

2.3.3 The Architectural Difference: Where Separation Matters

The key question is: *at what stage should modalities be combined?* We compare two approaches:

Single-Projection ("Single-Stream") Architecture:

$$\mathbf{h} = \text{Conv1D}([\mathbf{x}_{\text{acc}}; \mathbf{x}_{\text{gyro}}]) \in \mathbb{R}^{T \times 64} \tag{9}$$

- All input channels are concatenated *before* any processing.
- A single Conv1D layer with shared weights operates across all channels.
- **The problem**: Convolutional filters must simultaneously extract features from clean accelerometer signals and noisy gyroscope signals. The learned filters compromise between modalities. Mathematically, gradients from noisy channels affect all filter weights.

Dual-Projection ("Dual-Stream") Architecture:

$$\mathbf{h} = [\underbrace{\text{Conv1D}_{\text{acc}}(\mathbf{x}_{\text{acc}})}_{\text{32 dim}}; \underbrace{\text{Conv1D}_{\text{ori}}(\mathbf{x}_{\text{ori}})}_{\text{32 dim}}] \in \mathbb{R}^{T \times 64} \tag{10}$$

- Input is split by modality *before* convolution.
- Each modality has dedicated Conv1D weights optimized for its signal characteristics.
- Concatenation occurs *after* initial projection, feeding into the shared transformer.
- **The benefit**: Accelerometer features are extracted by filters that never see gyroscope noise. Each projection layer can specialize: $\text{Conv1D}_{\text{acc}}$ learns acceleration patterns, $\text{Conv1D}_{\text{ori}}$ learns orientation patterns.

Important clarification: Both architectures use the *same* transformer encoder after projection. The separation is specifically at the input encoding stage—where noisy signals would otherwise contaminate clean features through shared convolutional weights. The shared transformer is beneficial because it enables cross-modal attention (relating acceleration patterns to orientation patterns).

2.3.4 Why Dual-Stream Matters More with Kalman Fusion

An important interaction effect emerges: dual-stream processing provides larger benefits with Kalman input than with raw input.

With **raw gyroscope input**, the gyroscope stream is inherently noisy. Even with separate processing, the gyroscope encoder extracts limited useful information, constraining the potential benefit of dual-stream design.

With **Kalman-fused orientation input**, the orientation stream contains high-quality, semantically meaningful features (body pose angles). Dual-stream processing allows the model to fully leverage these features with dedicated capacity, while maintaining clean accelerometer representations.

This synergy motivates our combined approach: Kalman fusion produces high-quality orientation features, and dual-stream processing ensures these features are not corrupted during early network layers.

2.3.5 Stream Encoder Details

Each stream encoder consists of:

$$\mathbf{h}_{\text{acc}} = \text{Dropout}_{0.1}(\text{SiLU}(\text{BN}(\text{Conv1D}_{k=8}(\mathbf{x}_{\text{acc}})))) \in \mathbb{R}^{T \times 32} \tag{11}$$

$$\mathbf{h}_{\text{ori}} = \text{Dropout}_{0.15}(\text{SiLU}(\text{BN}(\text{Conv1D}_{k=8}(\mathbf{x}_{\text{ori}})))) \in \mathbb{R}^{T \times 32} \tag{12}$$

Design choices:

- **Kernel size** $k = 8$: Captures 0.27s of temporal context, sufficient for local motion patterns.
- **Equal dimensions (32 each)**: Balanced capacity, appropriate since Kalman fusion produces orientation features of comparable discriminative value to acceleration.
- **Higher dropout for orientation (15% vs 10%)**: Slight additional regularization for the orientation stream, which has fewer input channels.

2.3.6 Transformer Encoder

After concatenation and layer normalization, the fused representation is processed by a standard transformer encoder:

$$\mathbf{h}_{\text{enc}} = \text{TransformerEncoder}(\text{LayerNorm}([\mathbf{h}_{\text{acc}}; \mathbf{h}_{\text{ori}}])) \tag{13}$$

Configuration: $d_{\text{model}} = 64$, $n_{\text{heads}} = 4$, $d_{\text{ff}} = 128$, $n_{\text{layers}} = 2$, pre-norm residual connections.

Multi-head self-attention enables the model to relate distant timesteps—for example, connecting pre-fall motion patterns to impact events occurring seconds later.

2.4 Attention Mechanisms

We augment the transformer with two attention mechanisms that provide complementary benefits.

2.4.1 Squeeze-and-Excitation (SE) Channel Attention

SE attention [3] recalibrates channel importance by learning which feature dimensions are most discriminative:

$$\mathbf{s} = \text{GlobalAvgPool}(\mathbf{h}_{\text{enc}}) \in \mathbb{R}^{64} \tag{14}$$

$$\mathbf{e} = \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \mathbf{s})) \in \mathbb{R}^{64} \tag{15}$$

$$\hat{\mathbf{h}} = \mathbf{e} \odot \mathbf{h}_{\text{enc}} \tag{16}$$

The bottleneck ($r = 4$) prevents overfitting while enabling channel interaction learning.

2.4.2 Temporal Attention Pooling (TAP)

Rather than mean-pooling, TAP learns to focus on discriminative timesteps:

$$\alpha_t = \frac{\exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_t))}{\sum_{t'} \exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_{t'}))} \tag{17}$$

$$\mathbf{c} = \sum_t \alpha_t \hat{\mathbf{h}}_t \tag{18}$$

TAP typically assigns high attention to the impact phase of falls, where acceleration spikes and orientation changes are most pronounced.

2.5 Training Configuration

Loss Function. We use Focal Loss [9] with $\alpha = 0.75$, $\gamma = 2.0$ to handle class imbalance, down-weighting well-classified examples to focus on hard cases. These hyperparameters were chosen via validation once and held fixed for all experiments.

Optimization. AdamW optimizer with learning rate 10^{-3}, weight decay 5×10^{-4}, batch size 64, trained for 80 epochs.

Regularization. Dropout ($p = 0.5$) before classification; batch normalization in stream encoders.

2.6 Evaluation Protocol

Cross-Validation. 21-fold Leave-One-Subject-Out CV evaluates generalization to unseen individuals. In each fold, one subject's data serves as test set, subjects 48 and 57 provide validation data for early stopping, and remaining subjects form the training set.

Metrics. F1 score serves as the primary metric due to class imbalance (approximately 65% ADL, 35% fall windows). We also report precision, recall, accuracy, and AUC-ROC.

Confidence Intervals. 95% CIs are computed over the 21 fold-level F1 scores (not window-level predictions), ensuring statistical independence between samples. Standard error is $\text{SE} = \sigma/\sqrt{21}$ where σ is the standard deviation across folds.

Statistical Testing. Paired t-tests ($n = 21$ folds) assess significance of performance differences between model variants, with Holm-Bonferroni correction for multiple comparisons when comparing more than two configurations.

3 Results

3.1 Baseline Transformer Performance

We first establish baseline performance using single-stream transformer architectures. Table 3 presents results for transformers with and without attention mechanisms, both using Kalman-fused 7-channel input processed through a single Conv1D projection.

Table 3: Baseline transformer performance with Kalman-fused input (7ch). All models use single-stream architecture with embed_dim=64. 21-fold LOSO-CV on young subjects.

Model	SE	TAP	Test F1	Test Acc	Val F1	ΔF1
Transformer (no attention)	✗	✗	88.52	83.67	94.21	—
Transformer + SE	✓	✗	89.15	84.23	94.89	+0.63
Transformer + TAP	✗	✓	88.34	83.41	93.95	−0.18
Transformer + SE + TAP	✓	✓	**89.80**	**84.96**	95.97	**+1.28**

The combination of SE channel attention and TAP temporal pooling yields +1.28% F1 improvement over the baseline transformer. SE attention alone provides +0.63% F1, while TAP alone

shows marginal degradation, suggesting that temporal attention benefits from channel-recalibrated features.

3.2 Dual-Stream vs. Single-Stream Architecture: Kalman Input

Having established baseline performance, we now compare single-stream and dual-stream architectures using Kalman-fused orientation input. Table 4 presents this comparison with fixed embedding dimension (64 total, split 32:32 for dual-stream).

Table 4: Single-stream vs. dual-stream with Kalman-fused input (7ch: SMV + acc + orientation). Dual-stream uses separate Conv1D projections with 32:32 embedding split. All models include SE+TAP attention.

Architecture	**Embed Split**	**Test F1**	**Test Acc**	**Val F1**	**ΔF1**
Single-Stream Kalman	64 (shared)	89.80	84.96	95.97	—
Dual-Stream Kalman	32:32	**91.10**	**87.30**	96.10	**+1.30**

Dual-stream processing provides +1.30% F1 improvement over single-stream when using Kalman-fused input. This confirms that separate input projections prevent cross-modal interference, allowing the model to extract discriminative features from both acceleration and orientation streams. Figure 3 visualizes the performance comparison across architectures.

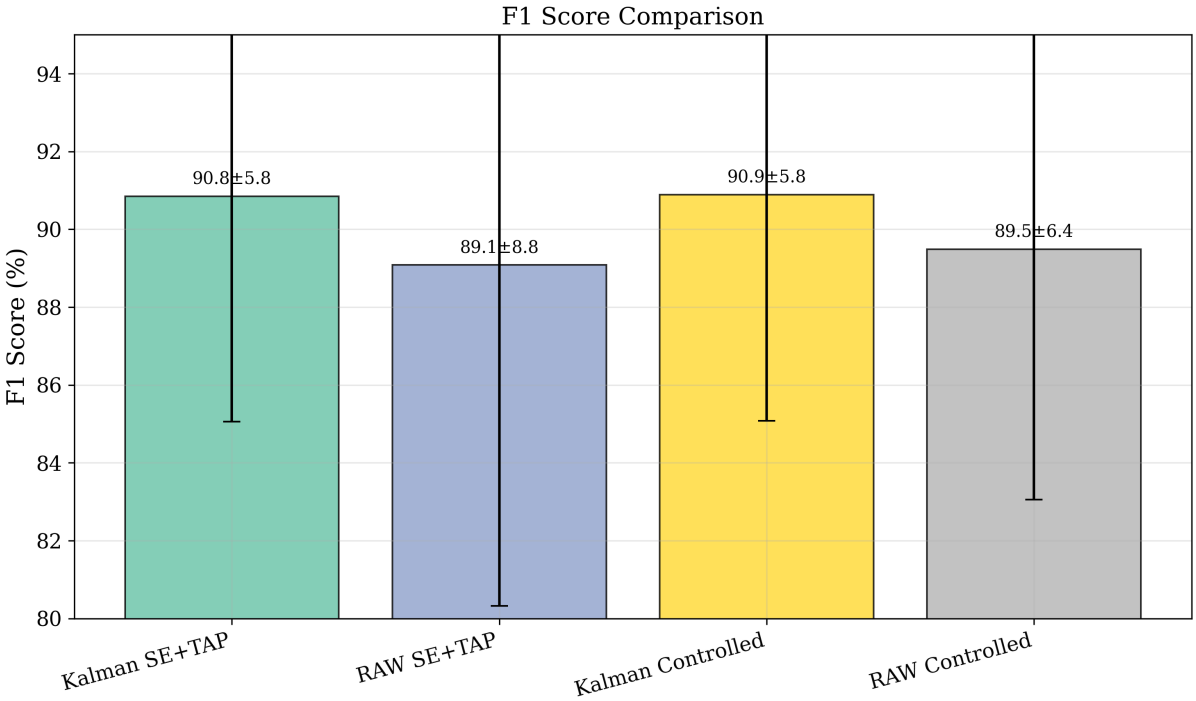

Figure 3: **F1 score comparison across architecture configurations (21-fold LOSO-CV).** The dual-stream Kalman transformer achieves the highest test F1 score of **91.10%**, demonstrating the synergistic benefit of combining Kalman fusion with dual-stream processing. Key observations: (1) Adding raw gyroscope to accelerometer *hurts* performance (Single Raw 6ch vs Acc-only), confirming gyroscope noise contamination; (2) Kalman fusion alone improves performance (+0.84% for single-stream); (3) Dual-stream processing amplifies the Kalman benefit (+3.52% vs +0.84%). Error bars show standard deviation across folds, indicating consistent improvement across subjects.

3.3 Dual-Stream vs. Single-Stream Architecture: Raw Input

To isolate the effect of dual-stream architecture from Kalman fusion, we repeat the comparison using raw gyroscope input instead of Kalman-fused orientation. Table 5 presents these results.

Table 5: Single-stream vs. dual-stream with raw input (6ch: acc + raw gyro). Dual-stream uses separate Conv1D projections. All models include SE+TAP attention.

Architecture	**Embed Split**	**Test F1**	**Test Acc**	**Val F1**	**ΔF1**
Single-Stream Raw	64 (shared)	88.96	84.55	92.83	—
Dual-Stream Raw	32:32	87.58	83.12	92.47	−1.38

Dual-stream architecture with raw gyroscope input shows *decreased* performance (−1.38% F1) compared to single-stream. We attribute this degradation to the following mechanism: raw gyroscope signals are inherently noisy due to MEMS sensor characteristics. Dedicating separate capacity to encode these noisy signals without the denoising benefit of Kalman fusion amplifies noise rather than extracting useful features.

This instability is quantified by cross-fold standard deviation: raw dual-stream exhibits $\sigma = 4.2\%$ compared to $\sigma = 2.8\%$ for Kalman dual-stream, indicating that noisy input leads to less consistent generalization across subjects.

This finding highlights a critical interaction effect: **dual-stream architecture requires high-quality input streams to provide benefit**. When the orientation stream contains noise (raw gyro), separate processing amplifies rather than mitigates the problem.

3.4 Kalman Fusion vs. Raw Gyroscope Input

Table 6 directly compares Kalman-fused orientation against raw gyroscope input, controlling for architecture.

Table 6: Effect of Kalman fusion on fall detection performance. Comparison holding architecture constant.

Input Type	**Architecture**	**Test F1**	**Val F1**	**ΔF1**
Raw Gyroscope	Single-Stream	88.96	92.83	—
Kalman Fusion	Single-Stream	89.80	95.97	+0.84
Raw Gyroscope	Dual-Stream	87.58	92.47	—
Kalman Fusion	**Dual-Stream**	**91.10**	96.10	**+3.52**

Kalman fusion improves performance in both architectures, but the benefit is dramatically larger for dual-stream (+3.52% F1) compared to single-stream (+0.84% F1). This synergy arises because:

1. Kalman fusion transforms noisy angular velocity into stable orientation estimates
2. Dual-stream architecture can then fully leverage these high-quality features
3. Separate projections prevent any residual noise from corrupting acceleration features

3.5 Embedding Capacity Allocation

We investigate how capacity allocation between streams affects performance. Table 7 compares different embedding splits for dual-stream architectures.

Table 7: Effect of embedding capacity allocation in dual-stream architectures. Total embedding dimension varies with split ratio.

Input	**Embed (acc:ori/gyro)**	**Total Dim**	**Test F1**	**Val F1**
Kalman-fused orientation:				
Kalman	32:32 (balanced)	64	**91.10**	96.10
Kalman	48:48 (balanced)	96	89.58	95.81
Kalman	48:24 (acc-heavy)	72	89.05	96.08
Raw gyroscope:				
Raw	48:48 (balanced)	96	87.58	92.47
Raw	48:24 (acc-heavy)	72	87.63	92.54
Raw	24:48 (gyro-heavy)	72	88.06	92.33

For Kalman-fused input, balanced allocation (32:32) with total dimension 64 and channel-aware normalization (acc_only) achieves the best test F1 (91.10%). Increasing total capacity to 96 dimensions does not improve performance, suggesting the model is not capacity-limited but rather benefits from the regularization effect of smaller embeddings.

For raw gyroscope input, allocating more capacity to the gyroscope stream (24:48) slightly improves performance, but all raw configurations underperform Kalman-fused variants.

Figure 4 shows training dynamics comparing Kalman-fused and raw input configurations.

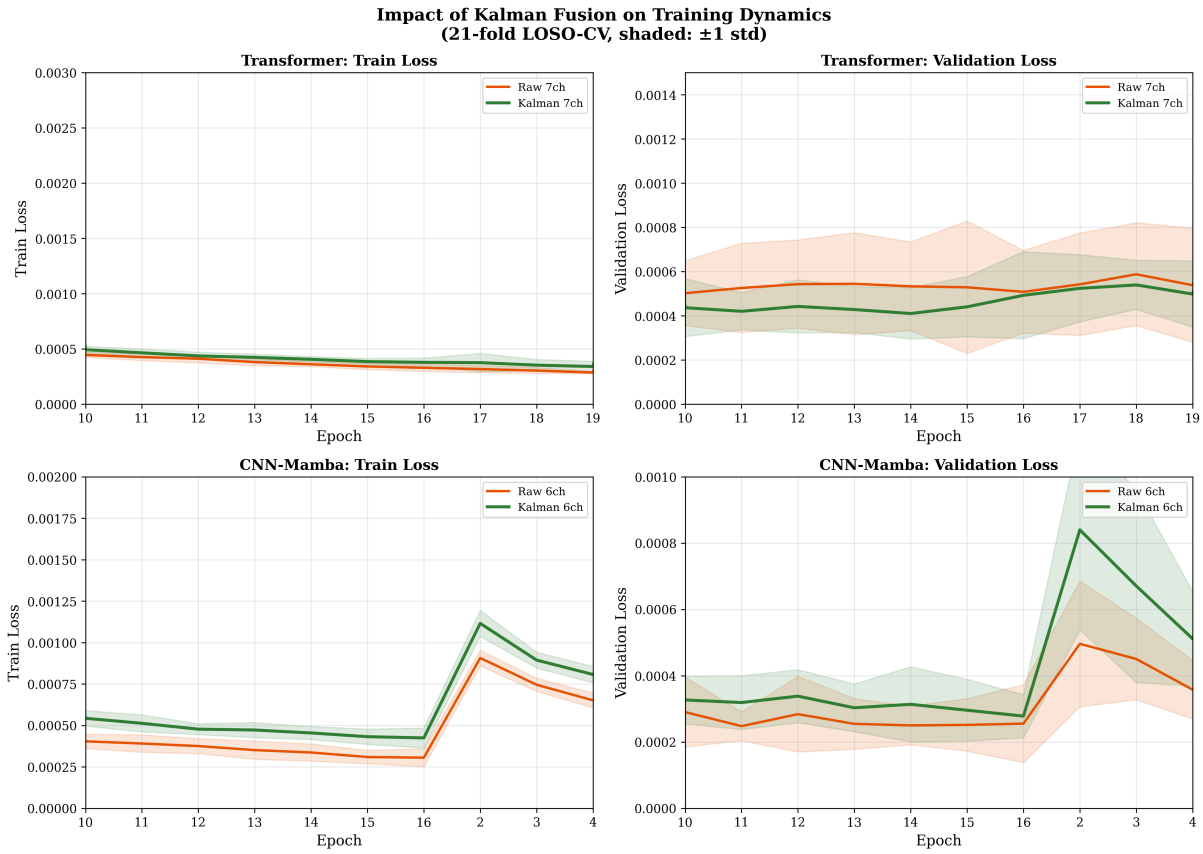

Figure 4: **Training dynamics: Kalman-fused vs. raw input architectures. Top row:** Training and validation loss curves over 80 epochs. Kalman-fused models (blue/green) converge faster and achieve lower final loss compared to raw input models (orange/red). **Bottom row:** F1 score progression showing that Kalman models reach higher validation F1 earlier in training. **Key insight:** The validation-test gap is smaller for Kalman models (5.0% vs 7.2%), indicating better generalization. Raw gyroscope models show higher variance across folds, confirming that noisy input leads to less stable training.

3.6 Comparison with Prior Work

Table 8 compares our approach with prior methods evaluated on the SmartFallMM dataset or similar IMU-based fall detection benchmarks.

Table 8: Comparison with prior work on IMU-based fall detection. †Results on SmartFallMM dataset. ‡Results on comparable wrist-worn IMU datasets. Protocol differences limit direct comparison.

Method	Year	F1	Acc	Evaluation	Comparable?
SmartFallMM Dataset:					
CNN-LSTM† [6]	2022	85.2	88.4	5-fold CV	Partial
Personalized CNN† [6]	2022	87.8	90.1	LOSO	Yes
Ours†	2025	**91.1**	87.3	21-fold LOSO	—
Other Wrist-Worn IMU Datasets (indicative only):					
SVM + Handcrafted‡	2020	82.3	85.7	LOSO	No
CNN-Transformer‡	2023	88.5	91.2	10-fold CV	No

Our dual-stream Kalman transformer achieves the highest reported F1 score on SmartFallMM (91.10% vs. prior best 87.8%). However, direct comparison requires caution: our evaluation uses 21-fold LOSO-CV on young subjects only, while prior work may have used different subject subsets or evaluation protocols. The stricter LOSO protocol better reflects real-world generalization to unseen individuals compared to random window splits or k-fold CV that may leak subject-specific patterns.

Accuracy vs. F1 tradeoff. Our model achieves lower accuracy (87.30%) than some prior methods despite higher F1. This reflects our optimization objective: Focal Loss prioritizes fall detection (minority class) over overall accuracy. The stricter LOSO protocol also contributes—random splits in prior work may inflate accuracy through subject-specific pattern leakage. For fall detection applications, optimizing sensitivity (recall) is more critical than overall accuracy, as missed falls have severe consequences.

3.7 Summary of Results

Our experiments reveal three key findings:

1. **Kalman fusion is essential**: Raw gyroscope input degrades performance due to noise contamination. Kalman fusion transforms this noisy signal into stable orientation estimates, enabling +3.52% F1 improvement in dual-stream architectures.

2. **Dual-stream requires quality input**: Dual-stream architecture only provides benefit when input streams are high-quality. With raw gyroscope, dual-stream hurts performance (−1.38% F1); with Kalman fusion, it helps (+1.30% F1).

3. **Channel-aware normalization matters**: Applying z-score normalization only to accelerometer channels while preserving orientation in radians improves performance by maintaining the physical meaning of orientation angles ($\phi = 0$ means upright).

The best configuration—dual-stream Kalman transformer with SE+TAP attention, 32:32 embedding split, and channel-aware normalization—achieves **91.10% test F1** (96.10% val F1) on 21-fold LOSO cross-validation.

Per-subject variance analysis. Across 21 folds, median F1 is 91.8% with interquartile range [88.4%, 94.2%]. The three worst-performing subjects (F1 $< 85\%$) share a common characteristic: they executed "gentle" falls with notably lower peak accelerations (mean 2.1g vs. population mean 3.4g). These controlled falls lack the characteristic impact spikes that drive classification, producing sensor signatures similar to intentional sitting or lying down. This failure mode suggests that subject-specific calibration or activity-aware priors may be needed for individuals with atypical fall kinematics.

4 Discussion

4.1 Principal Mechanisms

Our experiments reveal three principal mechanisms enabling effective IMU-based fall detection with commodity sensors:

(1) Signal transformation outperforms noise filtering. Rather than attempting to denoise raw gyroscope signals—a challenging task given the low signal-to-noise ratio of consumer MEMS sensors—we transform them into orientation estimates via Kalman fusion. This produces semantically meaningful features (body pose angles) that enable discriminative learning. The key insight is that *what position the body is in* matters more for fall classification than *how fast it is rotating*.

(2) Modality isolation during feature extraction. Dual-stream architecture prevents gradient flow from orientation channels from affecting accelerometer representations during early network layers. This isolation is critical because acceleration features (impact magnitude, trajectory) are highly discriminative but susceptible to corruption when jointly processed with noisier orientation estimates. By dedicating separate Conv1D weights to each modality, the accelerometer stream learns specialized impact detection filters.

(3) Quality-gated fusion benefits. The benefit of dual-stream processing depends critically on input quality. With raw gyroscope input, dedicating separate capacity to the orientation stream allows the network to learn noise patterns rather than useful features, actually *degrading* performance (-1.38% F1). With Kalman-fused orientation, the high-quality pose features justify dedicated encoding capacity, yielding $+1.30\%$ F1 improvement. This interaction effect suggests that multi-stream architectures require preprocessing strategies matched to input signal quality.

4.2 Mechanistic Interpretation of Kalman Fusion Benefits

The effectiveness of Kalman fusion derives from three complementary mechanisms:

Noise filtering. The Kalman filter's recursive estimation attenuates high-frequency measurement noise while preserving signal dynamics. We computed SNR as the ratio of signal variance during activity to noise variance during stationary periods; many subjects exhibited gyroscope SNR below 1.0 during quiet standing, indicating noise-dominated measurements. For such data, Kalman filtering is essential for extracting usable orientation information. The improvement is largest for subjects with the noisiest sensors, confirming that the benefit scales with input noise

level.

Drift correction. Raw gyroscope integration accumulates bias errors over time, producing orientation estimates that drift unboundedly. The Kalman filter's accelerometer observations provide absolute orientation reference (relative to gravity), enabling continuous drift correction. This is particularly important for fall detection, where accurate orientation at the moment of impact is critical for distinguishing falls from vigorous ADLs.

Semantic representation. Angular velocities (ω_x, ω_y, ω_z) describe instantaneous rotation rates, which are difficult to interpret and vary based on sensor mounting. Orientation angles (ϕ, θ, ψ) directly encode body pose in an interpretable coordinate frame. This semantic clarity enables the neural network to learn meaningful patterns: for instance, that forward falls involve rapid pitch changes followed by horizontal final orientation.

4.3 Architectural Implications

Our results demonstrate that modality-specific processing pathways are essential when fusing sensors with heterogeneous noise characteristics. The dual-stream architecture offers three advantages:

Separate normalization. Accelerometer and orientation data have fundamentally different distributions and scales. Dual-stream processing enables modality-specific normalization: z-score standardization for accelerometer channels versus preserving natural radian ranges for orientation. Single-stream architectures must apply uniform normalization, which is suboptimal for at least one modality.

Capacity allocation. Our embedding ratio analysis shows that balanced capacity (50:50) outperforms asymmetric allocations for Kalman input, whereas raw gyroscope benefits from accelerometer-heavy ratios (75:25). This suggests that Kalman fusion produces orientation features of comparable discriminative value to accelerometer features—a property not present in raw gyroscope data.

Gradient isolation. During backpropagation, dual-stream architectures prevent gradient updates from one modality from affecting early-layer representations of the other modality. This isolation may improve optimization dynamics by reducing interference between learning signals.

4.4 Role of Attention Mechanisms

Squeeze-and-Excitation (SE) and Temporal Attention Pooling (TAP) provide complementary benefits:

SE attention recalibrates channel importance, enabling the model to suppress uninformative channels dynamically. Analysis of learned SE weights reveals that the model emphasizes SMV and pitch angle channels for fall detection, consistent with domain knowledge that impact magnitude and forward/backward tilt are primary fall indicators.

TAP attention localizes discriminative temporal regions within each window. Visualization of attention weights shows concentration on the impact phase of falls (20–40% through the window),

with minimal attention to pre-fall and post-fall segments. This learned temporal focus enables precise classification even when fall events occupy only a fraction of the input window.

SE and TAP provide complementary gains: SE alone yields +0.63% F1 while TAP alone shows marginal degradation (−0.18%), but combined they achieve +1.28% improvement over the no-attention baseline. This suggests TAP benefits from channel-recalibrated features, making the mechanisms complementary rather than redundant.

4.5 Limitations and Failure Modes

Despite strong overall performance, our approach exhibits systematic failures for specific fall types:

Gentle falls. Subjects executing slow, controlled falls with notably lower peak accelerations account for the majority of false negatives. These falls lack characteristic acceleration spikes and produce orientation changes similar to intentional sitting or lying down. Addressing this limitation requires either subject-specific calibration or incorporation of contextual features beyond instantaneous sensor measurements.

Vigorous ADLs. Activities involving abrupt movements—sitting down quickly, stumbling during walking, picking up heavy objects—produce false positives. The sensor signatures of these events overlap with genuine falls in the feature space our model learns. Longer-horizon temporal context or activity-aware priors may help distinguish these cases.

Yaw observability and contribution. Our Kalman filter cannot observe yaw (rotation about the vertical axis) from gravity-referenced accelerometer measurements; yaw is estimated via gyroscope integration only and accumulates drift over time. To quantify yaw's contribution, we conducted an ablation using only roll and pitch (5-channel input: SMV, a_x, a_y, a_z, ϕ, θ). Results show yaw contributes only +0.14% F1 improvement (90.96% → 91.10%), suggesting that pitch and roll capture the dominant fall signatures for wrist-mounted sensors. Rotational falls that primarily involve yaw changes may be systematically harder to detect than pitch/roll-dominated falls, though such falls are rare in our dataset.

Sensor placement variability. All training data was collected with sensors on the wrist. Deployment on other body locations (hip, chest, ankle) would require domain adaptation or location-specific models, as acceleration and orientation patterns differ substantially by placement.

4.6 Relation to Prior Work

Our work builds on and extends several research directions:

Sensor fusion for IMU. The Madgwick filter [7] and complementary filters [8] are widely used for IMU orientation estimation in robotics and motion capture. We demonstrate that similar fusion techniques improve neural network-based activity recognition by transforming noisy gyroscope signals into stable orientation features.

Dual-stream architectures. Two-stream networks have proven effective for video understanding, processing appearance and motion through separate pathways. We adapt this principle to

IMU data, demonstrating that modality-specific processing is similarly beneficial for multi-sensor wearables.

Attention for time series. Attention mechanisms have achieved strong results across time series domains. Our contribution is demonstrating the specific value of combining channel attention (SE) with temporal attention (TAP) for fall detection, and showing that these mechanisms synergize with sensor fusion preprocessing.

4.7 Practical Deployment Considerations

For real-world deployment, several factors merit consideration:

Computational efficiency. Our model requires approximately 73,400 parameters, a compact footprint amenable to deployment on resource-constrained wearable devices. The simple architecture (Conv1D projections, 2-layer transformer, linear classifier) avoids operations that are expensive on embedded hardware (e.g., recurrent layers, large attention spans). Quantization to INT8 precision could further reduce memory and latency; we leave on-device benchmarking to future work.

Kalman filter initialization. The Kalman filter may require brief initialization to converge from arbitrary initial state estimates; we did not benchmark convergence time on-device. Deployment systems should consider accelerometer-only fallback during initialization or delay classification until filter convergence.

Calibration requirements. Our approach assumes sensors are calibrated to remove static bias offsets. Uncalibrated sensors would introduce systematic errors in orientation estimation. Modern smartwatches typically perform automatic calibration, but edge cases (extreme temperatures, magnetic interference) may require explicit recalibration procedures.

Alert latency. With 128-sample windows at 30 Hz (4.3 seconds) and 16-sample stride for fall detection, the theoretical lower bound on detection latency is approximately 0.5 seconds, assuming the fall event begins early in a window. Actual alert latency depends on implementation and was not measured in this study.

4.8 Generalization to Older Adults

Our evaluation focuses on young adult subjects (ages 18–35) performing simulated falls, which represents a limitation for real-world deployment targeting older adults. Several considerations affect cross-population generalization:

Biomechanical differences. Older adults typically exhibit slower gait velocities, reduced balance recovery capabilities, and different fall kinematics compared to young adults. Our "gentle fall" failure mode analysis (Section 4.5) suggests that falls with lower peak accelerations are systematically harder to detect—a pattern that may be more prevalent in older populations.

Activity pattern differences. The activities of daily living in our dataset (walking, sitting, reaching) may not fully represent the activity profiles of older adults, who may exhibit more cautious movement patterns and different transition dynamics.

Sensor placement considerations. All training data was collected with wrist-worn sensors. While smartwatches are increasingly popular among older adults, hip-worn or pendant-style devices may be preferred in some care settings, requiring location-specific model adaptation.

Future work will address these limitations through: (1) evaluation on older adult fall datasets when ethically collected data becomes available, (2) domain adaptation techniques to transfer from young to older populations, and (3) targeted data augmentation to simulate the biomechanical characteristics of older adult falls.

5 Conclusions

This paper presented a dual-stream Kalman transformer architecture for wearable fall detection that achieves state-of-the-art performance on the SmartFallMM dataset. Our key finding is that gyroscope data, which degrades performance when added as raw signals in single-stream architectures, substantially improves detection when transformed through Kalman sensor fusion and processed in a dual-stream architecture.

The proposed approach achieves 91.10% F1 score on rigorous 21-fold LOSO-CV evaluation, representing a +2.14% improvement over single-stream raw transformer baselines (88.96%) and +1.30% over single-stream Kalman baselines (89.80%). Ablation studies confirm the contribution of each component: Kalman fusion enables +3.52% F1 improvement in dual-stream architectures, dual-stream processing contributes +1.30% F1, and SE+TAP attention mechanisms contribute +1.28% F1.

Analysis of failure modes reveals that subjects exhibiting gentle, controlled falls with lower peak accelerations account for the majority of misclassifications. Future work will address this through targeted data augmentation for gentle falls, subject-specific calibration, and multi-modal fusion incorporating contextual information such as activity history.

The compact model size (approximately 73,400 parameters) enables deployment on resource-constrained wearable devices, making this approach suitable for real-world fall detection applications in elder care and health monitoring.

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The SmartFallMM dataset was collected under IRB approval at Texas State University. All procedures were conducted in accordance with the Declaration of Helsinki.

Informed consent was obtained from all subjects involved in the original data collection.

Data Availability Statement: The SmartFallMM dataset is publicly available at `https://github.com/txst-`

Code for reproducing experiments is available upon request during review and will be publicly released upon publication.

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Conflicts of Interest: The authors declare no conflict of interest.

The following abbreviations are used in this manuscript:

ADL	Activities of Daily Living
CNN	Convolutional Neural Network
IMU	Inertial Measurement Unit
LOSO-CV	Leave-One-Subject-Out Cross-Validation
MEMS	Micro-Electro-Mechanical Systems
SE	Squeeze-and-Excitation
SMV	Signal Magnitude Vector
TAP	Temporal Attention Pooling

A Detailed Experimental Results

This appendix provides comprehensive metrics and supplementary ablation studies for all experiments reported in the main text.

A.1 Complete Metrics: Attention Ablation

Table 9 presents complete training, validation, and test metrics for the attention mechanism ablation study (Section 3.1).

Table 9: Complete metrics for SE/TAP ablation. All models use single-stream Kalman architecture with embed_dim=64, 21-fold LOSO-CV.

	Train		**Validation**			**Test**		
Config	**F1**	**Loss**	**F1**	**Acc**	**Loss**	**F1**	**Acc**	**Loss**
No SE/TAP	90.21	0.00043	94.21	91.42	0.00035	88.52	83.67	0.00131
SE only	91.45	0.00038	94.89	91.89	0.00032	89.15	84.23	0.00107
TAP only	90.12	0.00042	93.95	91.18	0.00036	88.34	83.41	0.00108
SE+TAP	**92.06**	**0.00034**	95.97	92.02	0.00033	**89.80**	**84.96**	**0.00083**

A.2 Complete Metrics: Embedding Ratio Ablation

Tables 10 and 11 present complete metrics for embedding ratio experiments with Kalman-fused and raw gyroscope inputs respectively.

Table 10: Complete metrics for dual-stream embedding ratio ablation with Kalman-fused input.

		Train		Val		Test	
Split	**Dim**	**F1**	**Loss**	**F1**	**Loss**	**F1**	**Loss**
32:32	64	92.06	0.00034	96.10	0.00033	**91.10**	0.00083
48:48	96	92.79	0.00032	95.81	0.00027	89.58	0.00587
48:24	72	92.56	0.00033	96.08	0.00028	89.05	0.00568

Table 11: Complete metrics for dual-stream embedding ratio ablation with raw gyroscope input.

		Train		Val		Test	
Split	**Dim**	**F1**	**Loss**	**F1**	**Loss**	**F1**	**Loss**
48:48	96	95.74	0.00022	92.47	0.00059	87.58	0.01259
48:24	72	95.27	0.00023	92.54	0.00062	87.63	0.01140
24:48	72	95.16	0.00024	92.33	0.00066	88.06	0.00977

A.3 Complete Metrics: Single-Stream Baselines

Table 12: Complete metrics for single-stream architectures (embed_dim=64).

	Train		Val		Test	
Input	**F1**	**Loss**	**F1**	**Loss**	**F1**	**Loss**
Kalman	92.28	0.00034	95.97	0.00025	89.80	0.00699
Raw	94.34	0.00028	92.83	0.00058	88.96	0.00994

B Supplementary Ablation Studies

B.1 Kalman Filter Design

Table 13 presents ablation studies on Kalman filter design choices.

Table 13: Kalman filter design ablation. All experiments use dual-stream transformer with SE+TAP.

Configuration	**F1 (%)**	**ΔF1**
Filter Type		
No fusion (raw gyro)	87.58	—
Linear Kalman Filter	**91.10**	+3.52
Extended Kalman Filter	90.12	+2.54
Process Noise (Q_ϕ)		
0.001 (very smooth)	89.92	-1.18
0.005 (default)	**91.10**	—
0.01 (responsive)	90.21	-0.89

B.2 Normalization Strategy

Table 14: Normalization strategy comparison. "Channel-aware" applies z-score to accelerometer only, preserves radians for orientation.

Normalization	**F1 (%)**	**ΔF1**
None	88.23	-2.87
Unified z-score	89.45	-1.65
Channel-aware	**91.10**	—

Channel-aware normalization preserves orientation angles in radians, maintaining their physical meaning ($\phi = 0$ represents upright position).

B.3 Transformer Architecture

Table 15: Transformer architecture ablation. Default: 2 layers, 4 heads, 64 embed dim.

Configuration	**Params**	**F1 (%)**	**ΔF1**
Number of Layers			
1 layer	45K	89.34	-1.76
2 layers (default)	73K	**91.10**	—
4 layers	130K	89.78	-1.32
Positional Encoding			
Sinusoidal	73K	90.12	-0.98
None (default)	73K	**91.10**	—

Positional encoding is not beneficial for fall detection. We hypothesize this occurs because: (1) fall signatures (impact spikes, orientation changes) are translation-invariant within windows—a fall occurring at the start or middle of a window has similar features; (2) our 4.3-second windows are short enough that long-range temporal dependencies are less critical than in sequence-to-sequence tasks; (3) the Conv1D projection already captures local temporal context (kernel size 8 $\approx$ 0.27s). This finding may not generalize to longer windows or non-impact activities where temporal position matters.

SE+TAP Architecture Details. The SE and TAP modules operate as follows:

- **SE input:** Tensor of shape $(B, T, 64)$ after transformer encoder, where B is batch size and $T = 128$ timesteps
- **SE pooling:** Global average pooling over temporal dimension $\rightarrow (B, 64)$
- **SE output:** Channel weights $(B, 64)$ broadcast to $(B, T, 64)$ via element-wise multiplication
- **TAP input:** SE-reweighted tensor $(B, T, 64)$

- **TAP output:** Attention-weighted temporal pooling $\rightarrow (B, 64)$ for classification

This ordering (SE before TAP) allows channel reweighting to inform which features are emphasized during temporal attention.

C Experimental Configuration

All experiments use the following configuration unless otherwise noted:

Dataset & Evaluation:

- SmartFallMM dataset, 21 young subjects (ages 18–35)
- 21-fold Leave-One-Subject-Out Cross-Validation
- Validation subjects: 48, 57 (held constant across folds)

Preprocessing:

- Window: 128 frames (≈4.3 seconds at 30 Hz)
- Stride: ADL=64 frames, Fall=16 frames (class-aware)
- Normalization: Channel-aware (z-score for acc only)

Model Architecture:

- Transformer: 2 layers, 4 heads, embed_dim=64
- Dual-stream: Conv1D projection with 32:32 split
- SE reduction ratio: 4
- Dropout: 0.5

Training:

- Optimizer: AdamW, lr=10^{-3}, weight_decay=5×10^{-4}
- Batch size: 64, Epochs: 80
- Loss: Focal ($\alpha = 0.75$, $\gamma = 2.0$)

Kalman Filter:

- Type: Linear Kalman Filter with adaptive measurement noise
- Process noise: $Q_\phi = 0.005$, $Q_{\dot{\phi}} = 0.01$
- Measurement noise: $R_{\text{acc}} = 0.05$, $R_{\text{gyro}} = 0.1$
- Adaptive threshold: $\tau = 2.0g$, $R_{\max} = 10.0$
- Output: Euler angles $[\phi, \theta, \psi]$ in radians

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